AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Minor Project (ETMN100)

WEEKLY PROGRESS REPORT (WPR) - 1

For the week commencing: 20/07/2026-26/07/2026

Enrollment Number : A41105223039

Student Name: Anubhav Anand

Program: B.Tech CSE

Faculty Guide's Name: Dr. Girish Paliwal, Professor

Project Title: WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

TARGETS SET FOR THE WEEK

- Finalize project topic based on real-world relevance, technical feasibility, and originality against existing tools.
- Research and evaluate candidate data sources for JJM water quality data covering Uttar Pradesh and Bihar.
- Define technical architecture, methodology, and a week-by-week roadmap for the project.

ACHIEVEMENTS FOR THE WEEK

- Finalized project topic through a structured evaluation process, screening candidate ideas against existing tools and products to ensure originality before selecting a data-driven water quality triage system as the final direction.
- Identified four candidate sources for JJM water quality data: AIKosh (access request raised, pending approval), National Water Data Portal / CKAN (identified, pending verification), data.gov.in (confirmed available for infrastructure/connection data), and WQMIS public reports (identified as a potential live data source).
- Defined the technical architecture and methodology: a Python/FastAPI/PostgreSQL/SQLAlchemy pipeline with a contamination severity scoring model based on BIS 10500 permissible limits, a recurrence/trend detection module, and a comparative analysis layer for Uttar Pradesh and Bihar.
- Prepared a week-by-week project roadmap for the full 82-day duration.

FUTURE WORK PLANS

- Follow up on AIKosh access request; attempt direct verification of National Water Data Portal (CKAN) and WQMIS as alternate data sources.
- Set up local development environment (Python virtual environment, PostgreSQL, Docker, Git repository, FastAPI project skeleton).
- Load and inspect sample data for at least one test district each in Uttar Pradesh and Bihar to assess real data quality and completeness.
- Finalize exact contamination parameters and BIS 10500 threshold values to be used in the scoring model.